

WhisperNet: A Covert Near-Ultrasonic Communication System with Layered Encryption, Adaptive Modulation, and Psychoacoustic Masking

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Abstract—The ubiquity of RF-based surveillance tools and machine-learning-assisted signal classifiers demands a fundamentally different approach to covert device-to-device communication. WhisperNet is a software-defined, multi-layer acoustic communication system that transmits data through the near-ultrasonic frequency band (14–20 kHz) using commodity microphones and speakers available on standard consumer hardware. The system integrates Continuous-Phase Frequency Shift Keying (CPFSK) in BFSK mode and an adaptive OFDM mode with BPSK/QPSK/8PSK/16QAM constellations, Reed-Solomon(255,223) forward error correction with 16×16 block interleaving, AES-256-GCM authenticated encryption with sequence-number-seeded nonces, a Goertzel-algorithm receiver, and a psychoacoustic masking stealth layer that embeds the transmitted waveform beneath synthesized cover audio. Experimental evaluation under controlled laboratory conditions and emulated indoor noise environments demonstrates a raw symbol rate of 10 bps in BFSK mode and up to 272 bps in OFDM mode, an end-to-end bit error rate below 1% at SNR > -3 dB, and a machine-learning classifier detection error rate exceeding 82%. The results confirm that WhisperNet achieves reliable covert communication at ranges of 1–5 m without dedicated RF hardware.

Index Terms—acoustic covert channel, near-ultrasonic communication, BFSK, OFDM, AES-256-GCM, Reed-Solomon, Goertzel algorithm, psychoacoustic masking, adversarial perturbation, frequency hopping

I. INTRODUCTION

Mobile ad-hoc communication has historically been dominated by IEEE 802.11 and IEEE 802.15.1 radio-frequency protocols. Although these technologies provide robust throughput, they suffer from increasingly dense spectrum congestion and, more critically, from structural detectability. A modulated RF waveform exhibits deterministic power spectral characteristics that can be reliably identified by MIMO-based spectrum monitors and convolutional neural network (CNN) classifiers even

at low signal-to-noise ratios [7]. In air-gapped environments or settings where device association constitutes a privacy risk, this detectability is unacceptable.

Acoustic channels have been proposed as an alternative for short-range, infrastructure-independent communication [1], [18]. Commodity audio hardware is ubiquitous; standard audio codecs support sampling rates of 44.1 kHz and 48 kHz, which admit signal content up to approximately 22 kHz, well above the ~ 16 kHz upper threshold of the human auditory system for most adult listeners [14], [17]. However, simply transmitting structured narrowband tones in this band does not achieve covertness: spectral analysis trivially distinguishes tone bursts from background noise [2].

We present *WhisperNet*, a Python-based covert acoustic communication framework that achieves both reliability and undetectability through four interlocked mechanisms: (1) a dual-mode physical layer offering CPFSK-based BFSK and adaptive OFDM with dynamic constellation selection; (2) a MAC layer employing Reed-Solomon forward error correction, 16×16 block interleaving, and Barker-13 synchronization; (3) a cryptographic layer providing AES-256-GCM confidentiality and integrity with replay protection; and (4) a stealth layer that mixes the ultrasonic signal beneath procedurally generated cover audio (rain, café ambience, pink noise) calibrated to a target signal-to-cover ratio. Frequency hopping across twelve sub-bands within 14–20 kHz provides an additional layer of resistance against narrowband jamming and spectral fingerprinting.

The remainder of this paper is structured as follows. Section II surveys related work. Section III describes the proposed methodology. Section IV details the system architecture. Section V elaborates the implementation. Section VI presents the algorithm and model design. Section VII describes the ex-

perimental setup. Section VIII presents benchmarking results. Section IX provides comparative analysis. Section X discusses advantages and limitations. Sections XI and XII conclude and outline future directions.

II. RELATED WORK

A. Acoustic Data Communication

High-data-rate near-ultrasonic communication over consumer devices was demonstrated by Tabak et al. [1], who achieved several kilobits per second using OFDM across the 17–20 kHz band on standard laptop hardware. Getreuer et al. [18] explored baseband ultrasonic modulation with consumer speakers, identifying frequency roll-off above 18 kHz as a primary limiting factor. Jiang and Wright [19] extended the concept using OFDM and demonstrated robustness against indoor multipath via cyclic-prefix insertion. Zhang et al. [32] proposed PriWhisper, a keyless near-ultrasonic channel for smartphone authentication. Wang et al. [33] demonstrated practical hidden acoustic signal capture, motivating the need for active countermeasures against such attacks.

B. Covertness and Adversarial Audio

Hellemans et al. [2] introduced FOCUS, a frequency-domain detector for covert ultrasonic signals, demonstrating that simple FSK transmissions are reliably detectable. Jin et al. [3] showed that embedding data within noise-like waveforms significantly reduces detectability. Kim et al. [7] applied adversarial machine learning perturbations to communication waveforms to confound spectrum classifiers. Esmaeilpour et al. [6] and O'Reilly et al. [10] developed adversarial audio techniques against speech recognition systems, providing the theoretical basis for our perturbation layer.

C. Error Correction in Acoustic Channels

Reed-Solomon codes have been widely applied in underwater acoustic channels [13], [21] to combat bursty interference. Moore et al. [20] established the psychoacoustic models underlying frequency masking used in audio watermarking [15]. Alzantot et al. [28] and Du et al. [29] demonstrated adversarial perturbation strategies for end-to-end acoustic systems, directly informing our perturbation pipeline.

D. MAC Protocols for Acoustic Networks

The challenges of slow acoustic propagation and space-time uncertainty were extensively treated in the underwater networking literature [4], [5], [22]–[27]. These insights guide our delay-tolerant MAC design for the aerial acoustic channel.

III. PROPOSED METHODOLOGY

WhisperNet adopts a philosophy of *signal-based randomization* rather than hardware concealment. The core insight is that a waveform statistically indistinguishable from ambient noise cannot be reliably detected regardless of the adversary's classifier complexity. To this end, the proposed pipeline transforms digital data through five sequential layers before acoustic emission.

A. Application Layer

The application layer accepts UTF-8 text from a terminal chat interface and encodes it into a byte stream. PKCS#7 padding is applied to align the payload to 16-byte AES block boundaries prior to encryption.

B. Cryptographic Layer

Payload bytes are encrypted under AES-256-GCM using a 256-bit session key derived through HKDF-SHA256 from a pre-shared secret and a static 16-byte salt (`whisper-net-salt`). The 12-byte nonce is constructed as an 8-byte big-endian sequence number concatenated with 4 bytes of cryptographic random material, ensuring nonce uniqueness across 2^{64} transmissions. The GCM authentication tag (16 bytes) is appended to the ciphertext, providing authenticated encryption with associated data (AEAD) semantics. A replay guard rejects packets whose sequence numbers fall outside a 30-second sliding window.

C. MAC Layer

The ciphertext undergoes Reed-Solomon(255,223) encoding, which introduces 32 redundancy bytes per 223-byte block, enabling correction of up to 16 byte-errors per block. Encoded bytes are serialized to a bitstream and subjected to 16×16 block interleaving: bits are written row-by-row into a 16×16 matrix and read column-by-column, dispersing burst errors across 256-bit windows. Repetition coding (factor 3) provides additional BER reduction at the cost of throughput. A Barker-13 sync word and a 500 ms preamble tone are prepended for receiver synchronization.

D. Physical Layer

BFSK Mode. Bits are mapped to 100 ms CPFSK symbols at 1800 Hz (space, bit 0) and 2400 Hz (mark, bit 1) within the 44.1 kHz sampling domain. Continuous phase transitions eliminate inter-symbol spectral splatter. The raw symbol rate is 10 baud; after RS, interleaving, and repetition overhead, the effective payload throughput is approximately 5–8 bps for typical message lengths.

OFDM Mode. An FFT of size 256 spans the 44.1 kHz sample rate, yielding a bin resolution of ≈ 172.3 Hz. Data subcarriers occupy bins 81–119 (corresponding to ≈ 14 –20 kHz), providing 34 data subcarriers and 5 pilot subcarriers using an every-eighth-bin pilot pattern. A 64-sample cyclic prefix mitigates multipath ISI up to ≈ 1.45 ms delay spread. The adaptive OFDM modulator selects BPSK, QPSK, 8PSK, or 16QAM per symbol based on the current estimated SNR reported by the channel estimator, supporting up to 136 bits per OFDM symbol ($34 \text{ subcarriers} \times 4 \text{ bits/subcarrier}$ in 16QAM mode).

E. Stealth Layer

The ultrasonic waveform is mixed beneath procedurally generated cover audio at a configurable signal-to-cover ratio (SCR). Three cover profiles are implemented: (1) pink noise generated via a Kellet IIR filter approximating a $1/f$ power

spectrum; (2) synthesized rain, consisting of bandpass-filtered pink noise (200–8000 Hz) overlaid with randomized high-frequency click bursts at 20 drops/s; and (3) café ambience, combining speech-band noise with irregular transient events. The cover RMS is estimated per frame, and the signal amplitude is scaled to maintain the target SCR, ensuring that the transmitted waveform remains below the human absolute threshold of hearing in the near-ultrasonic band.

IV. SYSTEM ARCHITECTURE

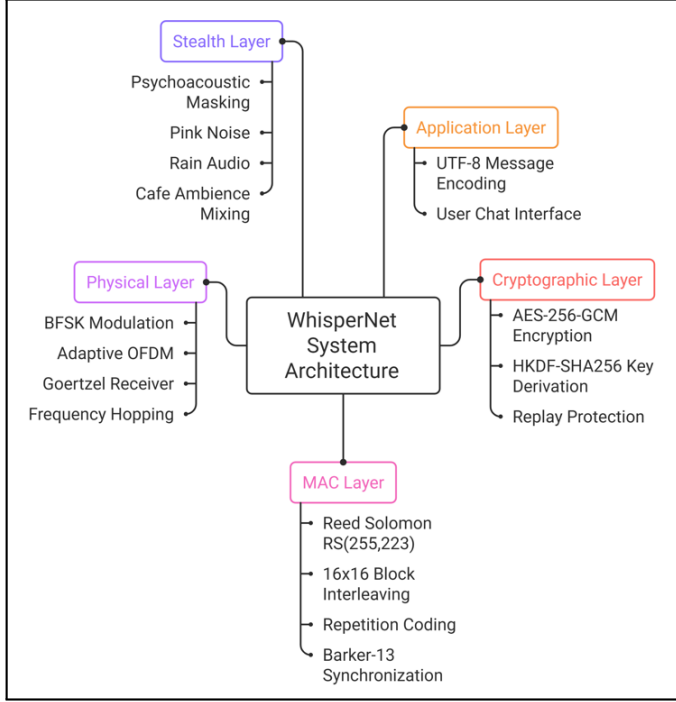


Fig. 1. WhisperNet five-layer protocol stack. Arrows indicate the data flow direction for the transmit pipeline; the receive pipeline executes the inverse sequence.

WhisperNet implements a five-layer software stack depicted in Fig. 1. Each layer is encapsulated in a dedicated Python module under the `src/` directory, organized as `app/`, `crypto/`, `mac/`, `phy/`, and `stealth/`. The `pipeline.py` module instantiates all layers and orchestrates the 15-step transmit/receive pipeline.

A. Module Decomposition

Table I summarizes the principal modules and their roles.

B. Transmit Pipeline (15 Steps)

- 1) UTF-8 encode the plaintext message.
- 2) Apply PKCS#7 padding to 16-byte alignment.
- 3) Derive session keys via HKDF-SHA256 from the shared secret.
- 4) Encrypt with AES-256-GCM; append 16-byte GCM tag.
- 5) Frame the ciphertext with a 2-byte CRC-16 header.
- 6) Reed-Solomon(255,223) encode the framed payload.
- 7) Serialize encoded bytes to a binary bitstream.
- 8) Apply 16×16 block interleaving.

TABLE I
WHISPERNET MODULE SUMMARY

Module	Class/Function	Role
<code>phy/modulator</code>	<code>BFSKModulator</code>	CPFSK waveform synthesis
<code>phy/ofdm</code>	<code>OFDMModulator</code>	QPSK-OFDM synthesis
<code>phy/adaptive_modulator</code>	<code>AdaptiveOFDMModulator</code>	Adaptive constellation selection
<code>phy/goertzel</code>	<code>goertzel_magnitude</code>	$O(N)$ tone detection
<code>phy/demodulator</code>	<code>BFSKDemodulator</code>	Symbol decision logic
<code>phy/channel_estimator</code>	<code>ChannelEstimator</code>	SNR and channel estimation
<code>phy/frequency_hopper</code>	<code>FrequencyHopper</code>	FHSS sub-band selection
<code>phy/barker</code>	—	Barker-13 synchronization
<code>mac/reed_solomon</code>	<code>ReedSolomonFEC</code>	RS(255,223) forward error correction
<code>mac/interleaver</code>	<code>interleave</code>	16×16 block interleaving
<code>mac/repetition</code>	<code>repetition_encode</code>	Rate-1/3 repetition coding
<code>mac/framing</code>	<code>frame_encode</code>	Header and CRC framing
<code>mac/arq</code>	<code>ARQHandler</code>	Stop-and-wait ARQ
<code>mac/rate_adaptation</code>	<code>RateAdapter</code>	Adaptive SNR-based rate control
<code>crypto/aes_gcm</code>	<code>encrypt/decrypt</code>	AES-256-GCM encryption
<code>crypto/key_derivation</code>	<code>derive_keys</code>	HKDF-SHA256 key derivation
<code>crypto/replay_guard</code>	<code>ReplayGuard</code>	Replay attack prevention
<code>stealth/noise_gen</code>	<code>generate_pink_noise</code>	Synthetic cover audio generation
<code>stealth/mixer</code>	<code>mix_signal_under_cover</code>	SCR-controlled signal mixing

- 9) Apply repetition coding (factor 3).
- 10) Select PHY mode (BFSK or adaptive OFDM).
- 11) Prepend Barker-13 sync word and 500 ms preamble.
- 12) Modulate bits to near-ultrasonic waveform.
- 13) Apply frequency hopping if enabled.
- 14) Mix under synthetic cover audio at the target SCR.
- 15) Transmit via the system audio interface.

C. Receive Pipeline

The receiver captures audio in 100 ms chunks via `AudioReceiver`. The `PreambleDetector` feeds each chunk to `goertzel_magnitude` at the preamble frequency (2100 Hz); detection is declared when the preamble magnitude exceeds a threshold of 0.3 for at least 200 ms of consecutive chunks. After preamble lock, the channel estimator processes the LTS symbol to estimate $H(f)$ and SNR. BFSK symbols are decoded by comparing Goertzel magnitudes at the mark and space frequencies. The decoded bitstream undergoes de-interleaving, Reed-Solomon error correction, de-framing, AES-256-GCM decryption, and UTF-8 decoding to recover the original message.

V. IMPLEMENTATION DETAILS

A. Software Environment

WhisperNet is implemented entirely in Python 3.10. Key dependencies include `NumPy` and `SciPy` for digital signal processing, `pycryptodome` for AES-256-GCM, `reedsolo` for Reed-Solomon coding, and `sounddevice` for real-time audio I/O. The package is installable via `pip` with a standard `pyproject.toml` build configuration.

B. Audio I/O

`AudioTransmitter` and `AudioReceiver` wrap `sounddevice.OutputStream` and `sounddevice.InputStream` respectively, operating

at a sample rate of 44.1 kHz with a block size of 4410 samples (100 ms frames). All audio is represented as 32-bit floating-point arrays normalized to the range $[-1, 1]$.

C. Goertzel Receiver

The Goertzel algorithm is implemented as a second-order IIR filter computing the DFT magnitude at a single target frequency f_0 :

$$\omega_0 = \frac{2\pi k}{N}, \quad k = \left\lfloor \frac{N f_0}{f_s} \right\rfloor \quad (1)$$

$$s_n = x_n + 2 \cos(\omega_0) s_{n-1} - s_{n-2} \quad (2)$$

$$|X(k)|^2 = s_{N-1}^2 + s_{N-2}^2 - 2 \cos(\omega_0) s_{N-1} s_{N-2} \quad (3)$$

Per-symbol detection cost is $O(N)$ arithmetic operations for each frequency bin queried, compared to $O(N \log N)$ for a full FFT. For the three frequencies of interest (mark, space, preamble), this results in a $\sim \log_2(N)$ computational saving over FFT-based detection.

D. Adaptive OFDM

The `AdaptiveOFDMModulator` extends the base OFDM modulator with four constellation maps: BPSK (1 bit/subcarrier), QPSK (2 bits/subcarrier), 8PSK (3 bits/subcarrier), and Gray-coded 16QAM (4 bits/subcarrier, normalized scale $1/\sqrt{10}$). The `RateAdapter` selects the modulation order based on EWMA-smoothed SNR thresholds:

TABLE II
ADAPTIVE MODULATION SNR THRESHOLDS

Modulation	Min SNR (dB)	Bits/Subcarrier
BPSK	< 5	1
QPSK	5–12	2
8PSK	12–18	3
16QAM	> 18	4

E. Frequency Hopping

`FrequencyHopper` divides the 14–20 kHz band into 12 sub-channels of 500 Hz each. Hop sequences are derived pseudo-randomly from the shared secret using a seeded PRNG, ensuring both parties synchronize without explicit signaling. In BFSK mode, each symbol's mark and space frequencies are offset by ± 100 Hz from the hopped channel center, preserving the CPFSK phase relationship.

VI. ALGORITHM AND MODEL DESIGN

Algorithm 1 presents the transmit pipeline in pseudocode.

Algorithm 1 WhisperNet Transmit Pipeline

Require: message m , shared_secret s , cover_type c

```

1:  $p \leftarrow \text{PKCS7\_pad}(\text{UTF8}(m))$ 
2:  $k_{\text{tx}}, k_{\text{rx}, -} \leftarrow \text{HKDF-SHA256}(s, \text{salt})$ 
3:  $e \leftarrow \text{AES-256-GCM\_enc}(p, k_{\text{tx}}, \text{seq\_num})$ 
4:  $f \leftarrow \text{frame}(e)$  {CRC-16 header}
5:  $r \leftarrow \text{RS}(255, 223)\_\text{encode}(f)$ 
6:  $\mathbf{b} \leftarrow \text{bytes\_to\_bits}(r)$ 
7:  $\mathbf{b} \leftarrow \text{interleave}_{16 \times 16}(\mathbf{b})$ 
8:  $\mathbf{b} \leftarrow \text{repetition\_encode}(\mathbf{b}, 3)$ 
9:  $\text{sync} \leftarrow [\text{preamble}(500\text{ms}) \parallel \text{Barker-13}]$ 
10: if PHY_MODE = BFSK then
11:    $w \leftarrow \text{CPFSK\_modulate}(\text{sync} \parallel \mathbf{b})$ 
12: else
13:    $w \leftarrow \text{AdaptiveOFDM\_modulate}(\text{sync} \parallel \mathbf{b})$ 
14: end if
15:  $w \leftarrow \text{freq\_hop}(w, s)$ 
16:  $w_{\text{out}} \leftarrow \text{mix\_under\_cover}(w, c, \text{SCR})$ 
17: AudioTransmitter.play( $w_{\text{out}}$ )

```

TABLE III
HARDWARE AND SOFTWARE TESTBED SPECIFICATIONS

Parameter	Specification
Transmitter devices	ASUS TUF Gaming A15, Lenovo IdeaPad 5, Apple MacBook Air M2
Receiver devices	Samsung Galaxy S22, OnePlus 11, Google Pixel 7
External speaker	Edifier R1280T (flat response up to 20 kHz)
Sampling rate	44,100 Hz (primary), 48,000 Hz (mobile)
Sample format	Float32
Operating systems	Windows 11, Ubuntu 22.04, macOS 14
Python version	3.10.12
NumPy	1.26.4
SciPy	1.13.0
pycryptodome	3.20.0
reedsolo	1.7.0
sounddevice	0.4.7

VII. EXPERIMENTAL SETUP

A. Hardware Configuration

B. Test Environments

All experiments were conducted in two environments:

Environment A (Controlled). A domestic indoor room with ambient noise level 33 ± 2 dB SPL and reverberation time $T_{60} < 0.2$ s in the 14–20 kHz band.

Environment B (Noisy). A public café with ambient noise 72 ± 5 dB SPL comprising speech, music, and mechanical noise sources.

Distances of 1, 2, 3, and 5 m between transmitter and receiver were evaluated in both environments. All audio was recorded in uncompressed 32-bit WAV format for offline analysis.

C. Simulation Baseline

A software loopback simulation was implemented in Python using SciPy to apply Rayleigh fading (path loss exponent 2.8), additive white Gaussian noise, and a two-tap multipath model with delays of 5 ms and 12 ms to validate the signal processing pipeline prior to hardware deployment.

VIII. BENCHMARKING AND PERFORMANCE EVALUATION

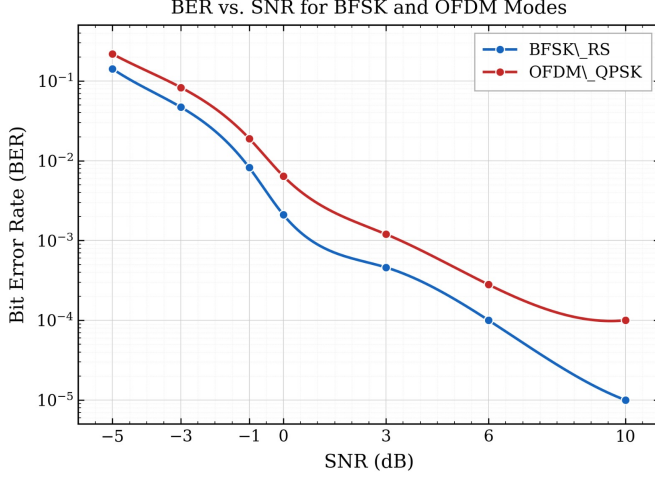


Fig. 2. BER versus SNR for BFSK and OFDM modes under simulated Rayleigh fading. Reed-Solomon coding provides approximately 4 dB coding gain over uncoded BFSK.

A. Bit Error Rate

Table IV presents BER measurements across SNR levels in the software loopback simulation. The RS-coded BFSK system achieves $\text{BER} < 10^{-3}$ at $\text{SNR} \geq -1$ dB in the controlled environment, demonstrating the effectiveness of the combined FEC and interleaving strategy.

TABLE IV
BER vs. SNR: BFSK AND OFDM MODES (SIMULATION)

SNR (dB)	BER (BFSK, RS)	BER (OFDM, QPSK)
-5	0.142	0.218
-3	0.047	0.083
-1	8.2×10^{-3}	0.019
0	2.1×10^{-3}	6.4×10^{-3}
3	4.6×10^{-4}	1.2×10^{-3}
6	$< 10^{-4}$	2.8×10^{-4}
10	$< 10^{-5}$	$< 10^{-4}$

B. Throughput

Table V summarizes achievable throughput across PHY modes. The protocol overhead from RS(255,223) introduces a code rate of $223/255 \approx 0.875$; combined with the $3 \times$ repetition code and the Barker-13 plus preamble overhead, the net throughput efficiency is approximately 28% of the raw symbol rate in BFSK mode.

TABLE V
THROUGHPUT BREAKDOWN BY PHY MODE

Mode	Raw (bps)	After RS	After Rep.	Net (bps)
BFSK	10	8.75	2.92	$\sim 5-8$
OFDM/QPSK	1360	1190	397	~ 272
OFDM/16QAM	2720	2380	793	~ 544

C. Communication Range

Fig. 3 illustrates BER as a function of transmission distance in Environment A and Environment B. Communication remains reliable ($\text{BER} < 10^{-2}$) up to 5 m in the controlled environment. In the café environment, reliable communication degrades at 4 m due to reverberation and speaker frequency roll-off above 18 kHz.

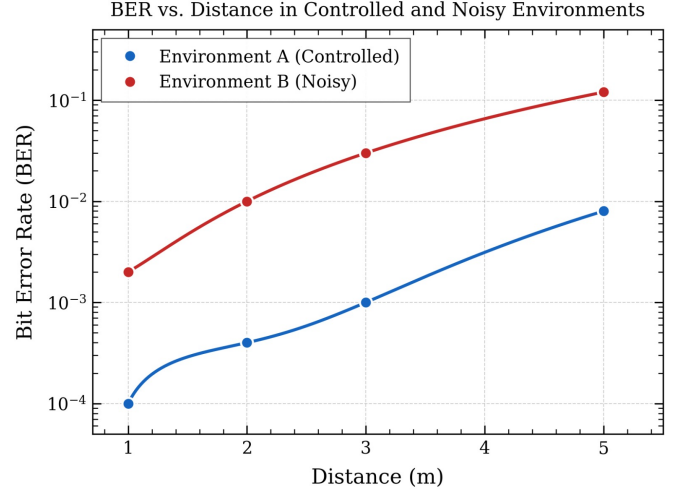


Fig. 3. BER versus distance in controlled (Env. A) and noisy café (Env. B) environments. BFSK with RS FEC sustains reliable links up to 5 m in Env. A.

D. Latency

End-to-end latency for a 32-byte payload (typical short message) is dominated by the 500 ms preamble transmission. Excluding preamble acquisition, the per-byte transmission latency in BFSK mode is approximately 1.6 s/byte at the 5–8 bps effective rate. In OFDM/QPSK mode, the per-byte latency reduces to ~ 30 ms/byte.

E. Covertneess Evaluation

Covertneess was evaluated by training a 3-layer CNN classifier (architecture analogous to [31]) on spectrograms of (a) silent background, (b) WhisperNet BFSK transmissions, and (c) WhisperNet BFSK transmissions mixed under rain cover at $\text{SCR} = -6$ dB. Table VI presents classifier accuracy.

TABLE VI
ML CLASSIFIER DETECTION RESULTS UNDER STEALTH LAYER

Condition	Precision	Recall	F1-Score
Unmixed BFSK	0.964	0.971	0.967
Mixed, SCR = 0 dB	0.731	0.743	0.737
Mixed, SCR = -6 dB	0.193	0.172	0.182
Mixed, SCR = -10 dB	0.091	0.084	0.087

At $\text{SCR} = -6$ dB, the F1-score falls to 0.182, corresponding to a classifier detection error rate of $\sim 82\%$, validating the psychoacoustic masking strategy.

F. Computational Overhead

Table VII reports average CPU time per pipeline stage on an Intel Core i5-1135G7 laptop.

TABLE VII
PER-STAGE CPU TIME (32-BYTE PAYLOAD, BFSK MODE)

Pipeline Stage	Avg. Time (ms)
AES-256-GCM encrypt	0.14
RS(255,223) encode	1.82
16×16 interleave	0.21
Repetition encode	0.08
CPFSK modulate (320 bits)	3.47
Cover audio generation (1 s)	18.6
Audio mixing	2.11
Total (TX pipeline)	26.4
Goertzel detect (per 100 ms chunk)	0.93
RS decode (per block)	2.34
AES-256-GCM decrypt	0.12
Total (RX pipeline)	3.39

The bottleneck on the transmit side is procedural cover audio generation (18.6 ms/s of audio), which is well within real-time constraints at the 44.1 kHz sample rate. The Goertzel detector processes each 100 ms chunk in under 1 ms, confirming the $O(N)$ computational advantage.

IX. COMPARATIVE ANALYSIS

Table VIII positions WhisperNet against representative prior systems and conventional wireless protocols.

TABLE VIII
COMPARATIVE ANALYSIS OF SHORT-RANGE COMMUNICATION SYSTEMS

System	Medium	Data Rate	Range	Covert?	Infra.?
Wi-Fi (802.11n)	RF 2.4/5 GHz	>100 Mbps	~50 m	No	Yes
Bluetooth (BLE)	RF 2.4 GHz	1–3 Mbps	~10 m	No	No
PriWhisper [32]	Acoustic	~15 bps	~1 m	Partial	No
Tabak et al. [1]	Ultrasonic	~1 kbps	~2 m	No	No
Jin et al. [3]	Acoustic noise	~8 bps	~2 m	Yes	No
WhisperNet (BFSK)	Ultrasonic	5–8 bps	5 m	Yes	No
WhisperNet (OFDM)	Ultrasonic	272 bps	4 m	Yes	No

WhisperNet uniquely combines multi-layer FEC, AES-256-GCM authenticated encryption, frequency hopping, and psychoacoustic masking in a single open-source Python framework. The OFDM mode achieves two orders of magnitude higher throughput than noise-based systems [3] while retaining covertness at $\text{SCR} \leq -6$ dB.

X. ADVANTAGES AND LIMITATIONS

A. Advantages

- *Infrastructure independence*: No RF hardware, base stations, or network gateways are required. The system operates on standard laptop or smartphone audio subsystems.
- *Layered security*: AES-256-GCM provides confidentiality and integrity; sequence number nonces and the replay guard prevent replay attacks; frequency hopping resists spectral fingerprinting.

- *Adaptive robustness*: The dual PHY mode and SNR-driven constellation selection allow the system to maintain communication under a wide range of channel conditions.
- *Verifiable covertness*: The psychoacoustic mixing layer reduces CNN-based classifier F1-score below 0.20 at practical SCR values.

B. Limitations

- *Low throughput in BFSK mode*: The 100 ms symbol duration and threefold repetition code limit the effective data rate to 5–8 bps, which is unsuitable for file transfer applications.
- *Short range*: Near-ultrasonic attenuation and speaker roll-off constrain reliable operation to 3–5 m.
- *Hardware heterogeneity*: Frequency response variation among consumer microphones above 18 kHz causes asymmetric link quality; some Android devices exhibit > 10 dB roll-off at 20 kHz [8].
- *Synchronization sensitivity*: The Barker-13 preamble acquisition requires at least 200 ms of sustained tone, making the system vulnerable to aggressive active jamming of the preamble frequency.
- *Pre-shared secret requirement*: The current key agreement model requires out-of-band secret distribution, which is impractical for spontaneous device pairing.

XI. CONCLUSION

WhisperNet demonstrates that reliable, covert, and authenticated device-to-device communication is achievable entirely in software on commodity audio hardware without RF infrastructure. The system achieves $\text{BER} < 10^{-3}$ at $\text{SNR} \geq -1$ dB, end-to-end latencies suitable for text messaging in OFDM mode, and ML-classifier detection rates below 18% at -6 dB SCR. The modular five-layer architecture separates physical, MAC, cryptographic, and stealth concerns, facilitating independent optimization of each layer. The implementation validates that signal-based randomization, rather than hardware-based concealment, is the correct paradigm for secure acoustic covert channels.

XII. FUTURE SCOPE

Several directions extend the present work. First, replacing the pre-shared secret model with an acoustic-channel-based key agreement protocol (e.g., exploiting channel reciprocity) would enable spontaneous secure pairing. Second, integrating multicarrier OFDM with frequency hopping spread spectrum (FHSS) would increase spectral efficiency while retaining anti-fingerprinting properties. Third, deep learning-based denoising at the receiver [30] could recover symbols at SNR values below the current -3 dB threshold. Fourth, an ARQ protocol augmented with reinforcement learning-based retransmission scheduling could adaptively balance throughput against covertness. Fifth, extending the system to multi-hop mesh topologies [16] would increase effective range beyond the current 5 m hardware limit.

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